

ARSH AUSIM MOBEEN

Mississauga, ON | 6474255174 | arshmobeen87817@gmail.com | arshmobeen.com | linkedin.com/in/arsh-mobeen | Canadian Citizen

EDUCATION

University of Western Ontario

London, ON

B.E.Sc. Software Engineering **GPA: 3.5**

Sep 2023 - 2027

- Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, Database Systems, Computer Networks, Systems Programming, Linux/ARM Assembly, Software Engineering, Machine Learning

PROFESSIONAL EXPERIENCE

Magna International / Modatek Systems

Milton, ON

AI Engineering Intern

May 2026 - August 2026

- Engineered **SQL-based data extraction workflows** to retrieve, validate, and structure engineering database records, enabling AI Agents to access operational data and automate technical support processes.
- Designed, trained, and connected multiple **Microsoft Copilot Studio AI Agents** using topics, actions, connectors, and grounded sources to automate Engineering and Operations workflows.
- Improved automation reliability through **Power Apps, intent mapping, fallback handling, iterative testing, and performance monitoring**, increasing workflow efficiency across engineering use cases.

Elite Life Financial

Mississauga, ON

Software Engineering Intern

November 2025 - March 2026

- Engineered **Python/JavaScript automation pipelines** using validation scripts and data transformation logic, reducing manual processing time by **30%+** while improving data accuracy across internal workflows.
- Engineered **Python validation tooling** and **SQL-backed reconciliation pipelines** to automate exception detection, reduce submission errors, and improve integrity across financial data workflows.
- Developed workflow automation scripts and internal process tooling using **backend scripting** and **data reconciliation logic**, increasing **operational throughput** and reducing repetitive manual tasks.

i3 Institute

Mississauga, ON

HR Technology Intern

June 2025 - October 2025

- Built a **full-stack workflow platform** using React, FastAPI, REST APIs, and dashboard-driven state tracking to automate onboarding workflows, improve task visibility, and increase cross-team process throughput.
- Engineered **automation pipelines and backend workflow tooling** for data ingestion, validation, routing, and status monitoring, reducing manual processing time by **30%+** while improving operational reliability.

PROJECTS

AbyssGPT | Python, Streamlit, Plotly, NetworkX, NLP

github.com/rayanamir1/AbyssGPT

- Developed an **AI-powered marine intelligence platform** using Python, Streamlit, Plotly and NetworkX, implementing graph-based pathfinding, risk-scoring algorithms and query routing for real-time exploration analysis across **multi-layer marine datasets**.
- Built a custom **NLP-driven query engine** using intent classification, geospatial modeling and multi-layer analytics for real-time route optimization, hazard assessment and intelligent decision support.

Personal Portfolio Website | React, HTML, CSS, JavaScript

github.com/Smokey878/Arsh-sPortfolio

- Developed and deployed a **responsive full-stack portfolio** using React, TypeScript, CSS and Vercel with modular architecture, dynamic animations and performance optimization.
- Engineered interactive UI features with **Framer Motion**, reusable React components and an integrated **OpenAI chatbot** to showcase projects through a high-performance front-end experience.

AI-Powered Traffic Light & Flood-Resilient Road Design | Arduino, Python, AutoCAD

- Engineered an **adaptive traffic control system** using Arduino IDE, Python control logic and real time water sensor-driven routing to reroute traffic under flooding conditions.
- Developed **rule-based signal optimization** and automated decision logic with embedded control algorithms, to improve traffic flow resilience, earning **Top Finalist / Excellence Award** recognition.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C++, C#, SQL, HTML/CSS, Bash, R

AI & ML: OpenAI API, Hugging Face Transformers, LangGraph, CrewAI, Prompt Engineering, ONNX, Multi-Agent Systems

Frameworks & Libraries: React, FastAPI, Node.js, Flask, Next.js, Vite, Tailwind CSS, Streamlit, NumPy, Pandas, Plotly, NetworkX, Scikit-learn, PyTorch

Cloud, Databases & Infrastructure: AWS, Docker, Linux, PostgreSQL, MySQL, Redis, Neo4j, Firebase, Git/GitHub, GitHub CI/CD, Postman, REST APIs, WebSockets, Modal

Developer Tools, Systems & Technologies: Data Structures & Algorithms, Object-Oriented Programming, Automation Pipelines, API Integration, Distributed Systems, OpenCV, Quartus, Intel FPGA

Leadership: President, Pakistani Students' Association | Electrical Engineer, Western Formula Racing | VP, Hemoglobal